

Course Code: CSE465	Course Title: Reinforcement Learning Type of Course: Theory Only	L-T-P-C	3 -0	0	3
Version No.	1.0				
Course Pre-requisites	Knowledge of programming in Python is required. Knowledge of probabilities/statistics, calculus and linear algebra is required. Machine learning background, as provided for example by COMP-551 or COMP-652 is required.				
Anti-requisites	NIL				
Course Description	The goal of this class is to provide an introduction to reinforcement learning, a very active research sub-field of machine learning. Reinforcement learning is concerned with building programs that learn how to predict and act in a stochastic environment, based on past experience. Applications of reinforcement learning range from classical control problems, such as power plant optimization or dynamical system control, to game playing, inventory control, and many other fields. Notably, reinforcement learning has also produced very compelling models of animal and human learning. During this course, we will study theoretical properties and practical applications of reinforcement learning. We will follow the second edition of the classic textbook by Sutton & Barto (available online for free, or from MIT Press), and supplement it as needed with papers and other materials.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Reinforcement Learning and attain Skill Development through Problem Solving Methodologies.				
Course Out Comes	On successful completion of the course the students shall be able to: Knowledge of basic and advanced reinforcement learning techniques. Identification of suitable learning tasks to which these learning techniques can be applied. Appreciation of some of the current limitations of reinforcement learning techniques. Formulation of decision problems, set up and run computational experiments, evaluation of results from experiments.				
Course Content:					
Module 1	Introduction	Assignment	Programming	No. of Classes:10	
Topics: Course logistics and overview. Origin and history of Reinforcement Learning research. Its connections with other related fields and with different branches of machine learning. Probability Primer Brush up of Probability concepts - Axioms of probability, concepts of random variables, PMF,					

PDFs, CDFs, Expectation. Concepts of joint and multiple random variables, joint, conditional and marginal distributions. Correlation and independence.

Module 2	Markov Decision Process	Assignment	Programming	No. of Classes:10
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Topics:

Introduction to RL terminology, Markov property, Markov chains, Markov reward process (MRP). Introduction to and proof of Bellman equations for MRPs along with proof of existence of solution to Bellman equations in MRP. Introduction to Markov decision process (MDP), state and action value functions, Bellman expectation equations, optimality of value functions and policies, Bellman optimality equations.

Module 3	Prediction and Control by Dynamic Programming	Assignment	Programming	No. of Classes:10
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Topics:

Overview of dynamic programming for MDP, definition and formulation of planning in MDPs, principle of optimality, iterative policy evaluation, policy iteration, value iteration, Banach fixed point theorem, proof of contraction mapping property of Bellman expectation and optimality operators, proof of convergence of policy evaluation and value iteration algorithms, DP extensions

Monte Carlo Methods for Model Free Prediction and Control
Overview of Monte Carlo methods for model free RL, First visit and every visit Monte Carlo, Monte Carlo control, On policy and off policy learning, Importance sampling.

Module 4	TD Methods and Policy Gradients	Assignment	Programming	No. of Classes:10
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Topics:

Incremental Monte Carlo Methods for Model Free Prediction, Overview TD(0), TD(1) and TD(λ), k-step estimators, unified view of DP, MC and TD evaluation methods, TD Control methods - SARSA, Q-Learning and their variants.

Getting started with policy gradient methods, Log-derivative trick, Naive REINFORCE algorithm, bias and variance in Reinforcement Learning, Reducing variance in policy gradient estimates, baselines, advantage function, actor-critic methods.

Targeted Application & Tools that can be used:

While Convolution Neural Network (CNN) and Recurrent Neural Network (RNN) are becoming more important for businesses due to their applications in Computer Vision (CV) and Natural Language Processing (NLP), Reinforcement Learning (RL) as a framework for computational neuroscience to model decision making process seems to be undervalued. Besides, there seems to be very little resources detailing how RL is applied in different industries. Despite the criticisms about RL's weaknesses, RL should never be neglected in the space of corporate research given its huge potentials in assisting decision making.

Tools: Torch, Google Colaboratory, Spider, Jupiter Notebook

Project work/Assignment:

This part is written for general readers. At the same time, it will be of greater value for readers with some knowledge about RL.

Resources management in computer clusters

Designing algorithms to allocate limited resources to different tasks is challenging and requires human-generated heuristics. The paper “Resource Management with Deep Reinforcement Learning” [2] showed how to use RL to automatically learn to allocate and schedule computer resources to waiting jobs, with the objective to minimize the average job slowdown.

State space was formulated as the current resources allocation and the resources profile of jobs. For action space, they used a trick to allow the agent to choose more than one action at each time step. Reward was the sum of $(-1/\text{duration of the job})$ over all the jobs in the system. Then they combined REINFORCE algorithm and baseline value to calculate the policy gradients and find the best policy parameters that give the probability distribution of actions to minimize the objective.

Traffic Light Control

Researchers tried to design a traffic light controller to solve the congestion problem. Tested only on simulated environment though, their methods showed superior results than traditional methods and shed a light on the potential uses of multi-agent RL in designing traffic system.

Five agents were put in the five-intersection traffic network, with a RL agent at the central intersection to control traffic signalling. The state was defined as eight-dimensional vector with each element representing the relative traffic flow of each lane. Eight choices were available to the agent, each representing a phase combination, and the reward function was defined as reduction in delay compared with previous time step. The authors used DQN to learn the Q value of the {state, action} pairs.

Robotics

There are tremendous works on applying RL in Robotics. Readers are referred to for a survey of RL in Robotics. In particular, trained a robot to learn policies to map raw video images to robot's actions. The RGB images were fed to a CNN and outputs were the motor torques. The RL component was the guided policy search to generate training data that came from its own state distribution.

Web System Configuration

There are more than 100 configurable parameters in a web system and the process of tuning the parameters requires a skilled operator and numerous trial-and-error tests. The paper “A Reinforcement Learning Approach to Online Web System Auto-configuration” showed the first attempt in the domain on how to do autonomic reconfiguration of parameters in multi-tier web systems in VM-based dynamic environments.

The reconfiguration process can be formulated as a finite MDP. The state space was the system configuration, action space was {increase, decrease, keep} for each parameter, and reward was defined as the difference between the given targeted response time and measured response time. The authors used the model-free Q-learning algorithm to do the task.

Text Book

"Reinforcement Learning: An Introduction", Richard S. Sutton and Andrew G. Barto, 2nd Edition

"Probability, Statistics, and Random Processes for Electrical Engineering", 3rd Edition, Alberto Leon-Garcia

"Machine Learning: A Probabilistic Perspective", Kevin P. Murphy

References

Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", Second Edition, MIT Press, 2019. Li, Yuxi. "Deep reinforcement learning." arXiv preprint arXiv:1810.06339 (2018). Wiering, Marco, and Martijn Van Otterlo. "Reinforcement learning." Adaptation, learning, and optimization 12 (2012):
E-Resources NPTEL course – https://onlinecourses.nptel.ac.in/noc19_cs55/preview https://archive.nptel.ac.in/courses/106/106/106106143/ https://www.digimat.in/nptel/courses/video/106106143/L35.html
Topics relevant to “SKILL DEVELOPMENT”: Real time Data Analysis using Reinforcement learning for Skill Development through Problem Solving techniques. This is attained through assessment component mentioned in course handout.